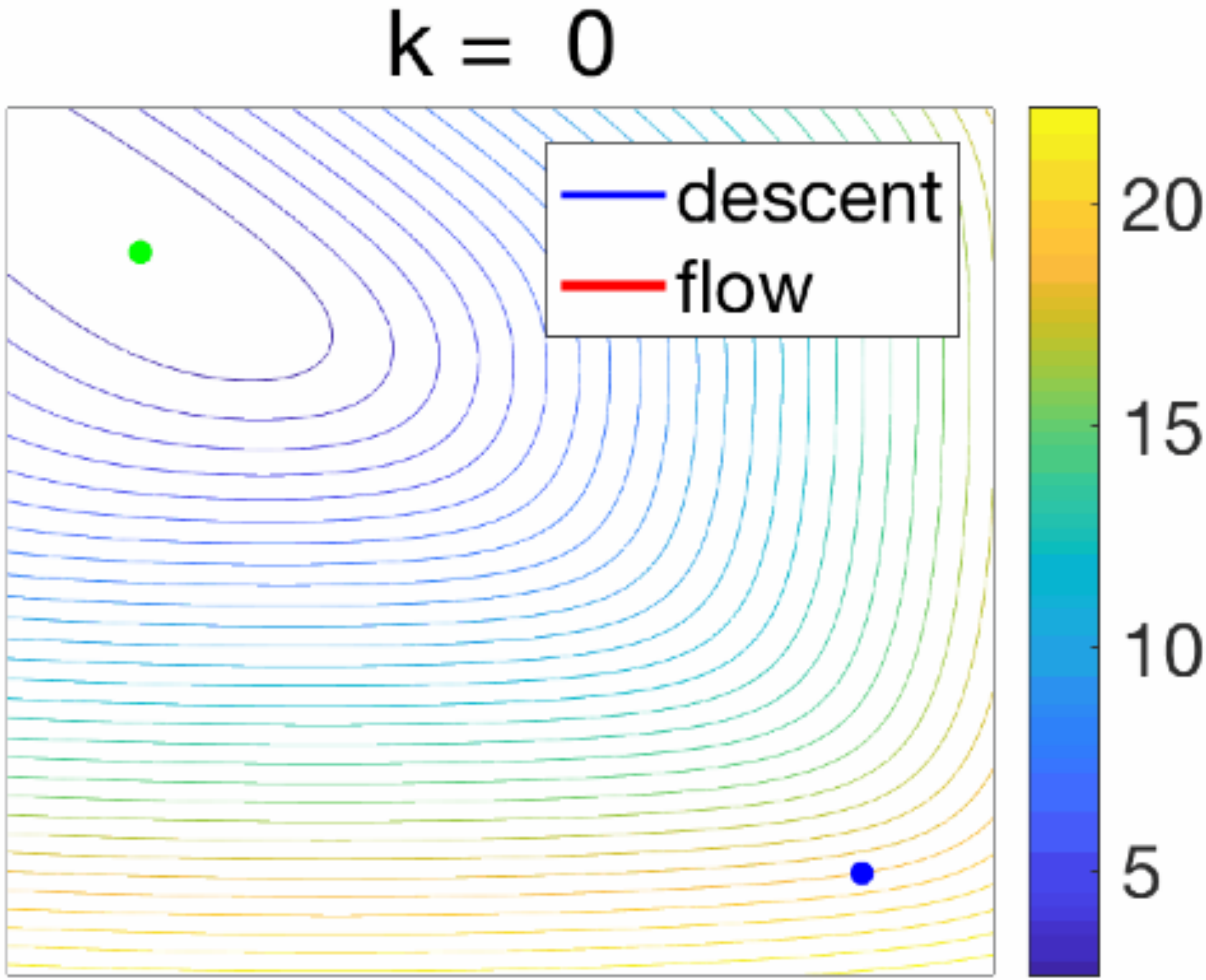
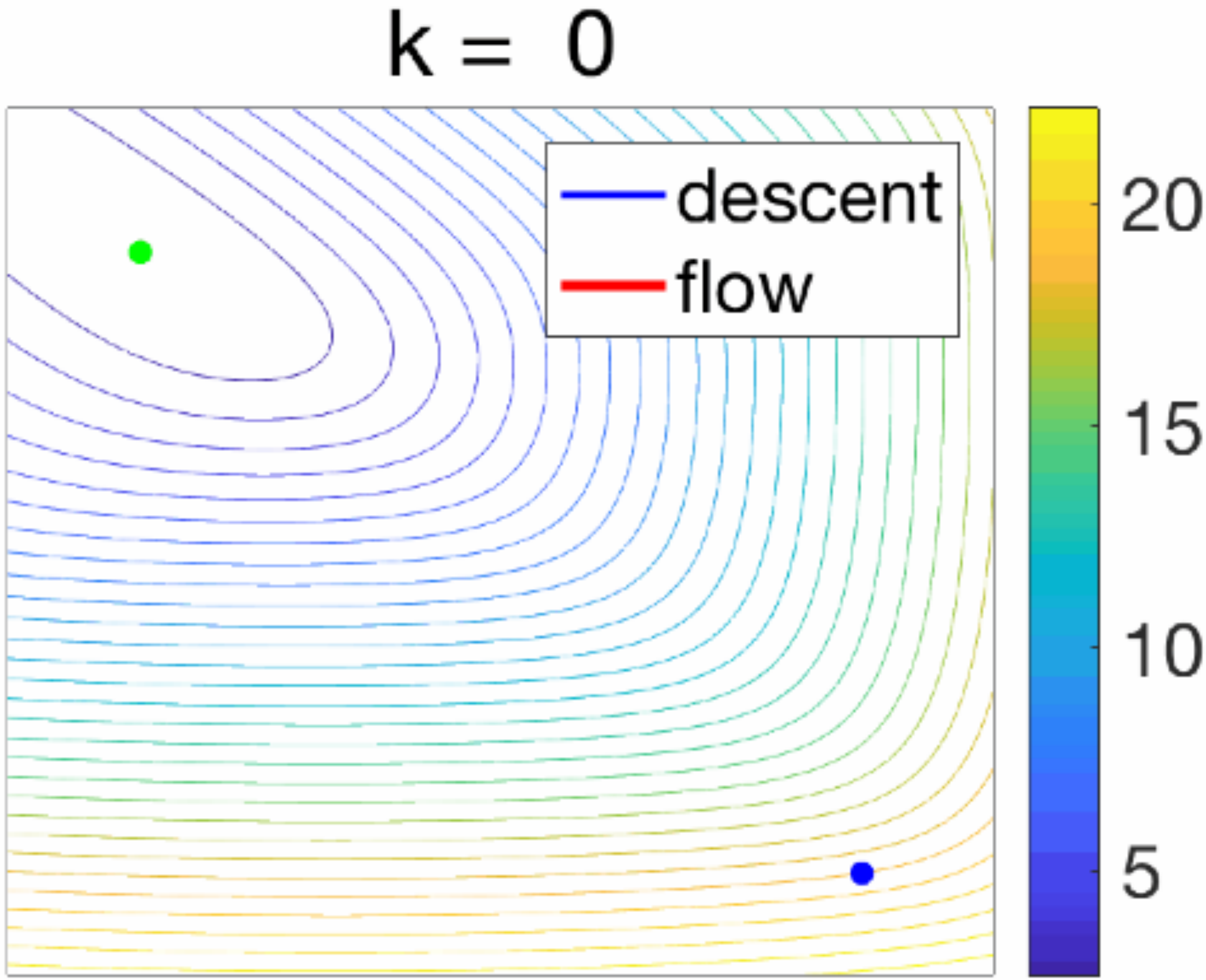
gradient flow: $\dot{x}(t) = -\nabla f(x(t))$

a curve $x(t)$ in $\mathbb{R}^d$

Practical algorithm requires time discretization


k = 0
descent
flow
20
15
10
5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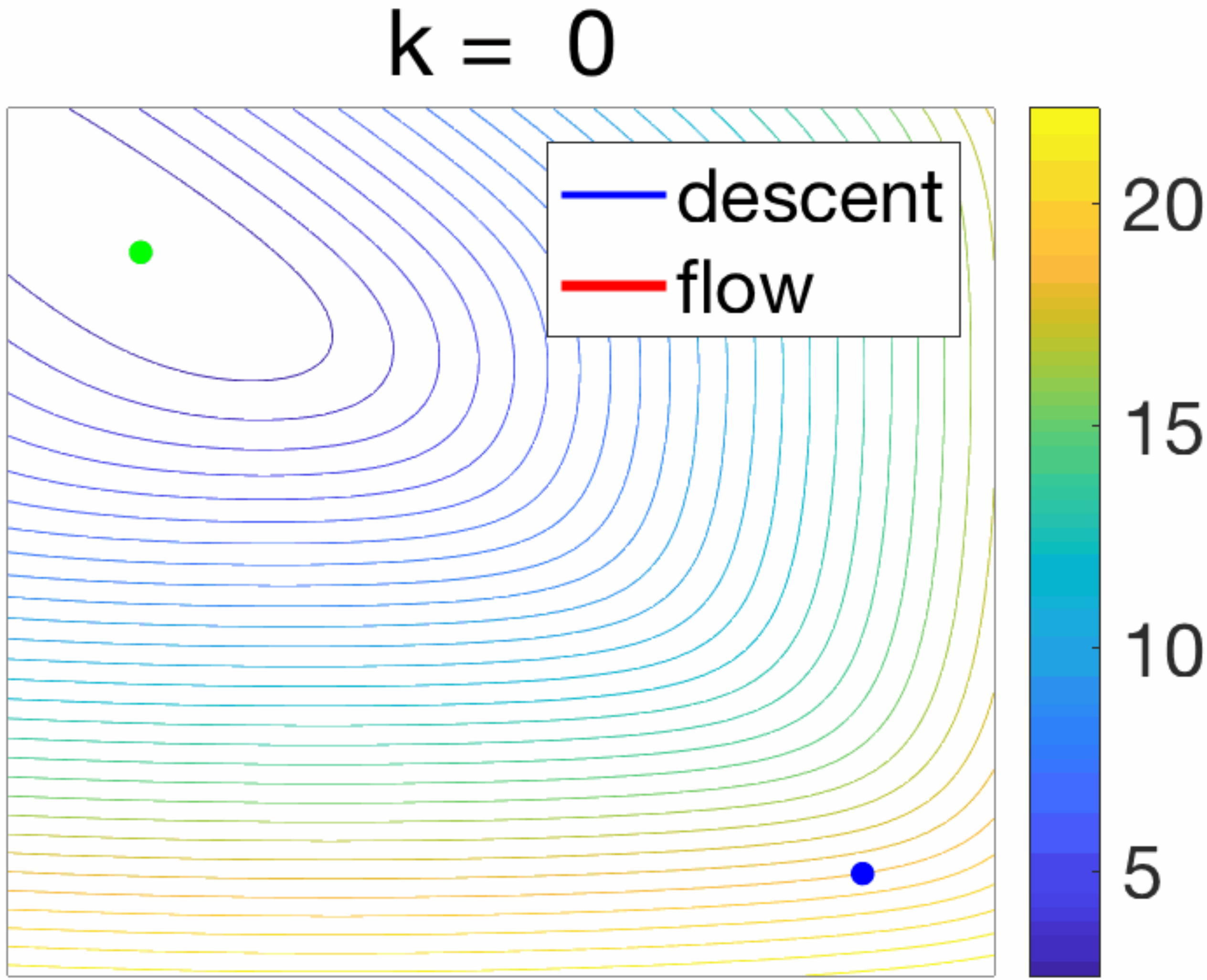
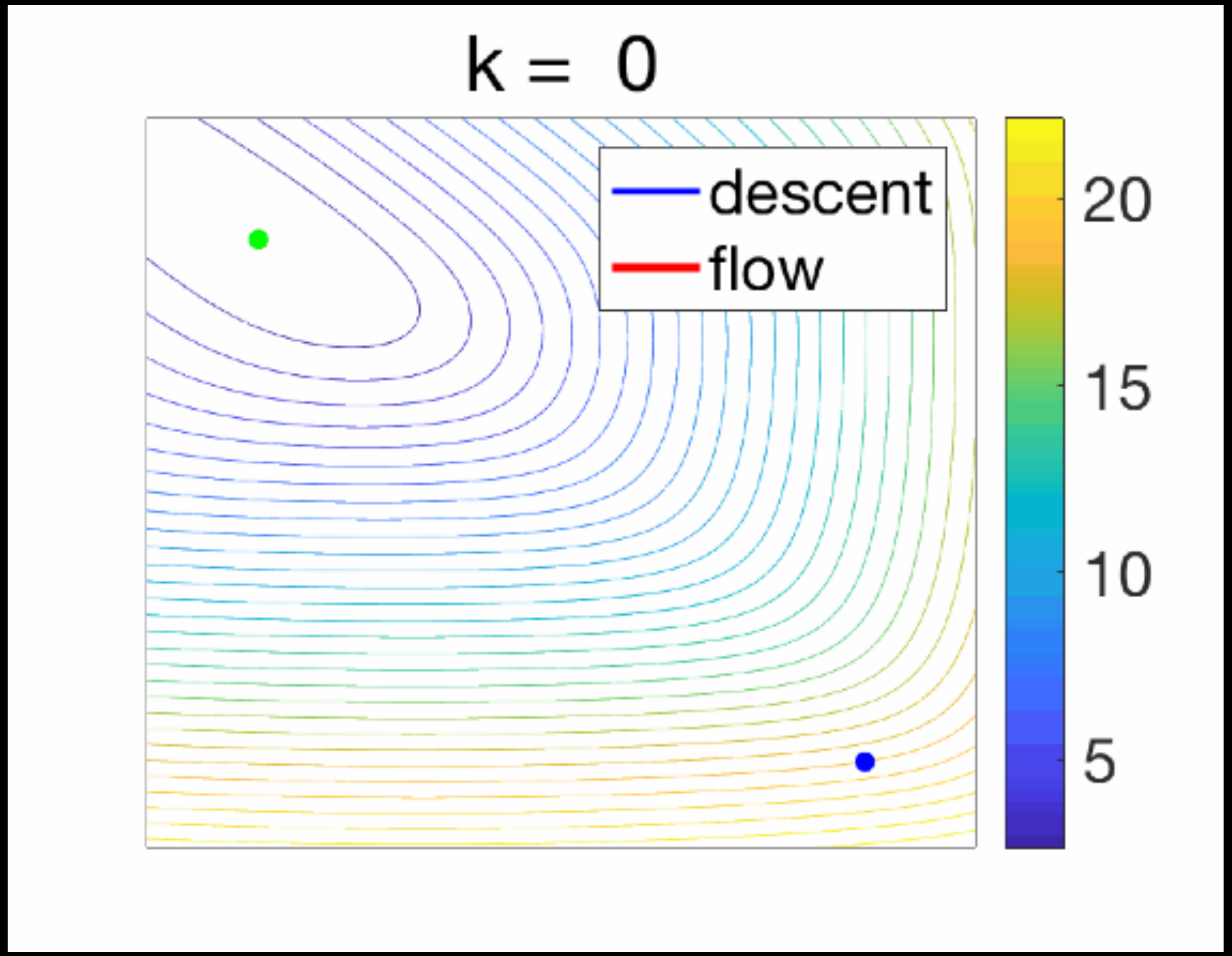

k = 0
descent
flow
20
15
10
5

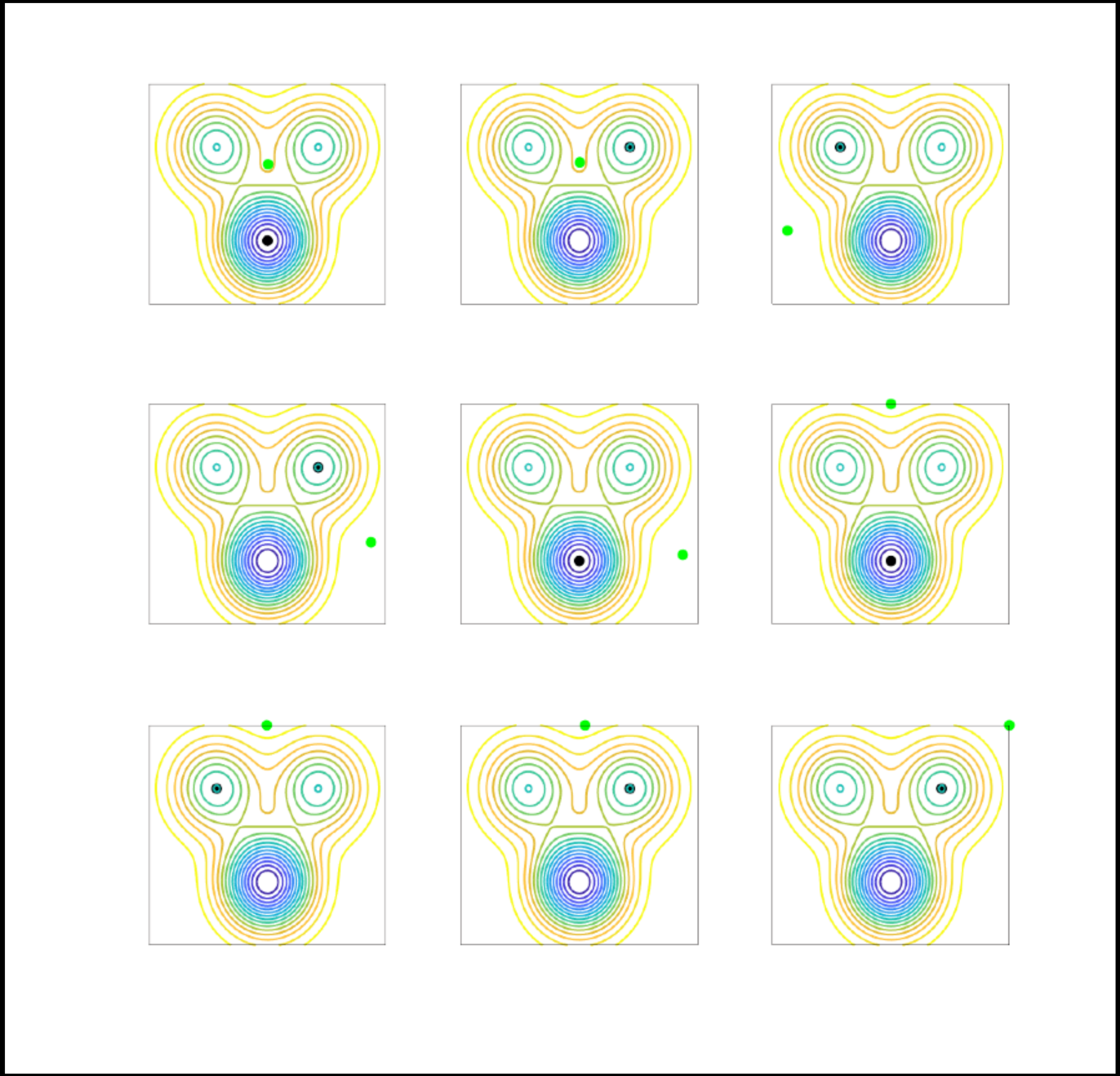
Warmup: Gradient Flows in $\mathbb{R}^d$

- Gradient descent: $x_{n+1} = x_n - \gamma \nabla f(x_n)$, $\gamma > 0$ stepsize
- Taking limit $\gamma \to 0$ leads to an ordinary differential equation called the **gradient flow:** $\dot{x}(t) = -\nabla f(x(t))$
- This is **a curve $x(t)$ in $\mathbb{R}^d$** that starts at $x(0) = x_0$ and moves by choosing at each instant the direction of steepest descent of f
- Pros:
  - Simplified analysis (no choice of step-size, etc)
  - Can often prove convergence of gradient flow in cases where proving things about gradient descent is hard
  - Connects to ODE/SDE theory
- But:
  - Cannot directly run flow as continuous-time! **Practical algorithm requires time discretization**



Credit: Francis Bach

Warmup: Gradient Flows in $\mathbb{R}^d$



Credit: Francis Bach